ModelIt!

Teacher's Guide

Can You Send a Message with Light?

The Science of Light Communication

Quick Reference

Grade Level	Grade 1
NGSS Standard	1-PS4-4
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: 1-PS4-4

Use tools and materials to design and build a device that uses light or sound to solve the problem of communicating over a distance.

Three-Dimensional Learning

Science Practice	**Designing Solutions** — Students design a device that uses light signals to communicate a message over a distance.
Disciplinary Core Idea	**PS4.C Information Technologies and Instrumentation** — People use devices to send and receive information, including light-based signals.
Crosscutting Concept	**Cause and Effect** — Students explore how changing signal brightness affects whether a message is successfully received.

CAST Testing Alignment

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

People have been using light to send messages for thousands of years. From ancient signal fires to modern fiber optic cables, light is one of the fastest and most reliable ways to communicate.

Light as Communication

Long before telephones or the internet, people used light to send messages over great distances. Ancient warriors lit fires on hilltops to warn of approaching enemies. Lighthouses flash patterns of light to guide ships safely past dangerous rocks. Traffic lights use red, yellow, and green to tell drivers what to do. Even today, the internet sends information as tiny pulses of light through thin glass cables called fiber optics. Light travels very fast, making it perfect for sending messages.

How Light Signals Work

For a light signal to work, three things must happen: someone must send the signal, the light must travel to the receiver, and the receiver must be able to see and understand it. Brighter lights can be seen from farther away. A code helps both people understand the message — like one flash means yes and two flashes mean no. If anything blocks the light, like a wall or fog, the message will not get through. That is why lighthouses are built very tall and use very bright lights.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L — Locate the System

Students identify signal brightness as the external component we control (how bright the flashlight is) and message received as the internal result (whether the partner gets the message).

E — Establish Relationships

Students discover that brighter signals are more likely to be received successfully — a positive relationship.

V — Visualize & Model

Students build a simple model showing how signal brightness connects to message received.

E — Evaluate Outcomes

Students test their model by trying bright and dim flashlights at different distances.

R — Revise & Extend

Students reflect on how their model explains why lighthouses need to be so bright and tall.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a "first draft" model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: "Based on what you just learned, would you add a component? Change a relationship?" Takes 10–15 minutes.
As a "what-if" tool	When students ask questions during class, say: "Great question — go test that in your model!" Students run new scenarios and report back.
For peer discussion	Partners compare models: "Why does your model have this arrow and mine doesn't?" Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience "what-if" challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"What if you needed to tell someone something but they were really far away and you could not yell? Could you use LIGHT?"*

Watch For

Show pictures of light signals: traffic lights, lighthouse, emergency vehicles. Ask: What is the light telling us?

Slide 2: What Will We Learn?

3 min

Say: *"Today we will learn how to use light to send messages. We will learn two words: SIGNAL and COMMUNICATE."*

Watch For

Practice communicating without words: thumbs up, waving, nodding. Say: These are all signals! Light can be a signal too.

Slide 3: Let's Be Scientists!

3 min

Say: *"How can we use light to send a message to someone far away? Has anyone ever played with flashlights at night?"*

Watch For

Students share ideas about how they could use light to say something. Record ideas on chart paper.

Slide 4: What Are the Parts?

5 min

Say: *"Our model has two parts: Signal Brightness is how strong the light is. Message Received is whether the other person sees and understands it."*

Watch For

Demonstrate with a flashlight: shine it bright, then cover it with a hand to make it dim. Ask: Which is easier to see?

Slide 5: How Are They Connected?

5 min

Say: *"When the signal is brighter, is the message easier or harder to receive? Brighter means BETTER communication!"*

Watch For

Students use sentence frame: When signal brightness goes up, message received goes up. Draw the connection arrow.

Slide 6: What Did We Find Out? 5 min

Say: *"We discovered that light can carry messages! Brighter signals work better, and people have used light signals for hundreds of years."*

Watch For

Review discoveries. Show one more example of light communication (fiber optic cable or Morse code lamp).

Slide 7: STEM Challenge 10 min

Say: *"Now work with a partner to create a light code! Decide what one flash, two flashes, and three flashes mean. Then test it!"*

Watch For

Distribute flashlights. Partners move apart and practice sending coded messages. Groups share their codes with the class.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Signal Brightness
- **INTERNAL:** Message Received

Relationships

- **Signal Brightness to Message Received:** POSITIVE (+) — A brighter signal is easier to see from far away, so the message is more likely to be received and understood. Dim signals are hard to see and the message may be lost.

Key Discoveries

- Light can be used to send messages to people far away
- Brighter signals are easier to see and understand from a distance
- The person receiving the message must be able to see the light for communication to work
- People have used light signals for hundreds of years, like lighthouses and signal fires

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"You can only communicate by talking"	People communicate in many ways besides talking! Sign language uses hands, Braille uses bumps for reading, traffic lights use colors, and lighthouses use light beams. Light signals are a powerful way to send messages, especially over long distances where voices cannot reach.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Light signals only work at night"	Light signals work during the day too, but they need to be very bright. Traffic lights work perfectly during the day because they use powerful LEDs. At night, even a small flashlight can be seen from far away because the dark background makes the light stand out more.